

UML Design of

ML Project

Student Placement Analysis
& Prediction

TEAM MEMBERS

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Academic Year

Why UML Design Matters



System Clarity

Apply UML modeling to visually represent the architecture, flow, and structure of an ML-based student placement prediction system.



ML Integration

Bridge Software Engineering practices with Machine Learning by designing standardized diagrams for classification and regression models.



Standardized Design

Enable team collaboration and communication through formally structured use-case, class, sequence, and activity diagrams.

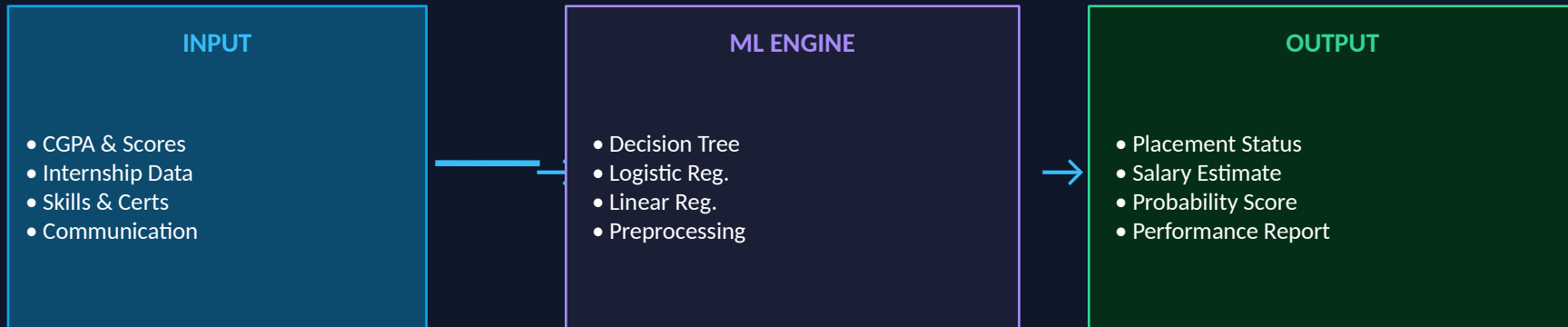
Problem Statement & Flow

Problem

Traditional placement assessment lacks predictive power. Students and institutions need data-driven placement forecasting.

Solution

ML-based classification (placement status) and regression (salary) with UML-designed system architecture for transparency.



Input Variables & Target Labels

CGPA

Cumulative Grade Point Average (0–10)

Internships

Number of internships completed

Coding Skill

Score from coding assessments

Communication

Verbal & written skill rating

Projects

No. of academic/personal projects

Certifications

Professional certificates earned

TARGET LABELS:

Placement Status (Placed / Not Placed) — Classification Task

Expected Salary (in LPA) — Regression Task

Algorithms & Their Roles

1

Decision Tree

Classification

Splits data into branches using feature thresholds to classify placement outcomes.

Predict: Placed / Not Placed

2

Logistic Regression

Classification

Estimates probability of placement using sigmoid function on weighted features.

Predict: Placed / Not Placed

3

Linear Regression

Regression

Fits a linear equation to continuous target variable (salary) using feature weights.

Predict: Expected Salary (LPA)

Preparing Data for ML Models

01

Data Cleaning

Handle missing values via mean/mode imputation. Remove duplicate records and fix inconsistent entries.

02

Outlier Removal (IQR)

Use Interquartile Range method to detect and remove statistical outliers from CGPA and salary columns.

03

Label Encoding

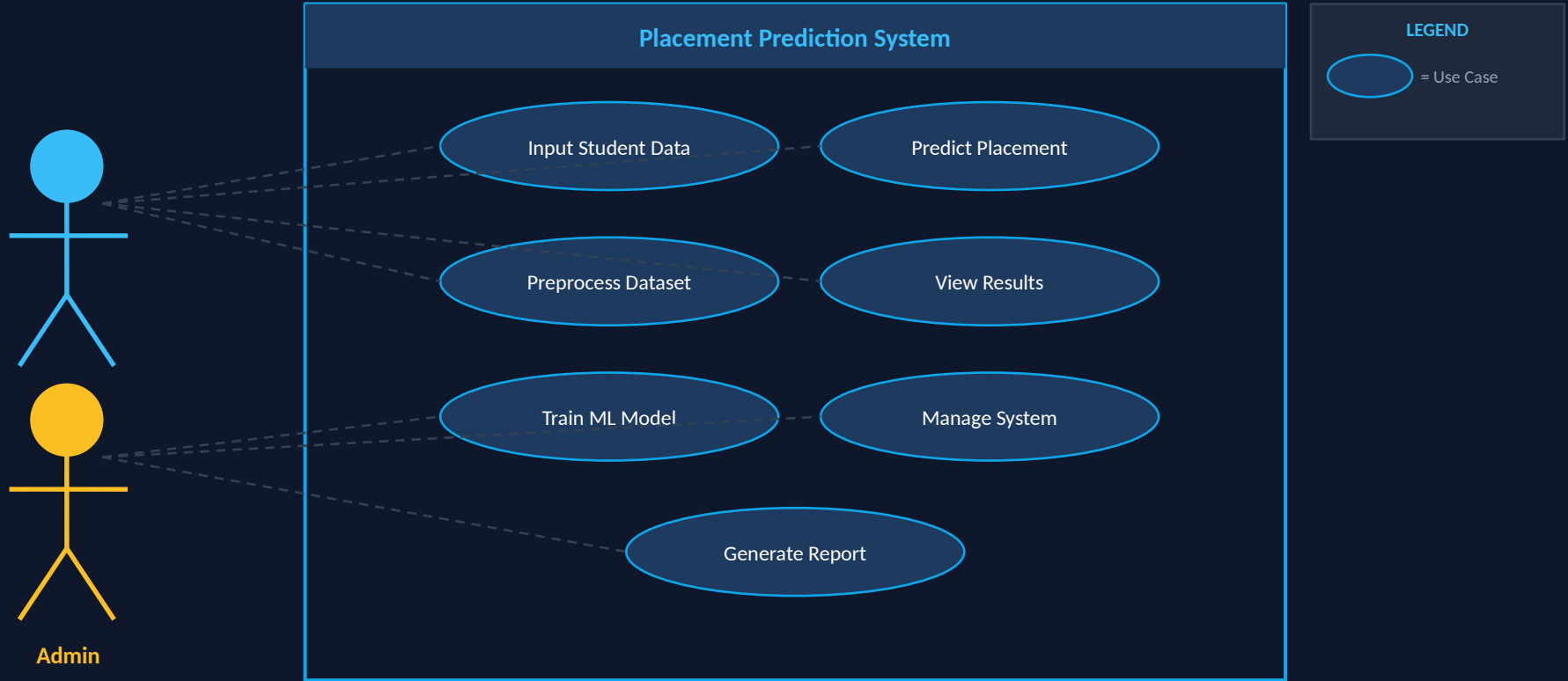
Convert categorical variables (Placement: Yes/No, Gender, Stream) to numeric format using sklearn LabelEncoder.

04

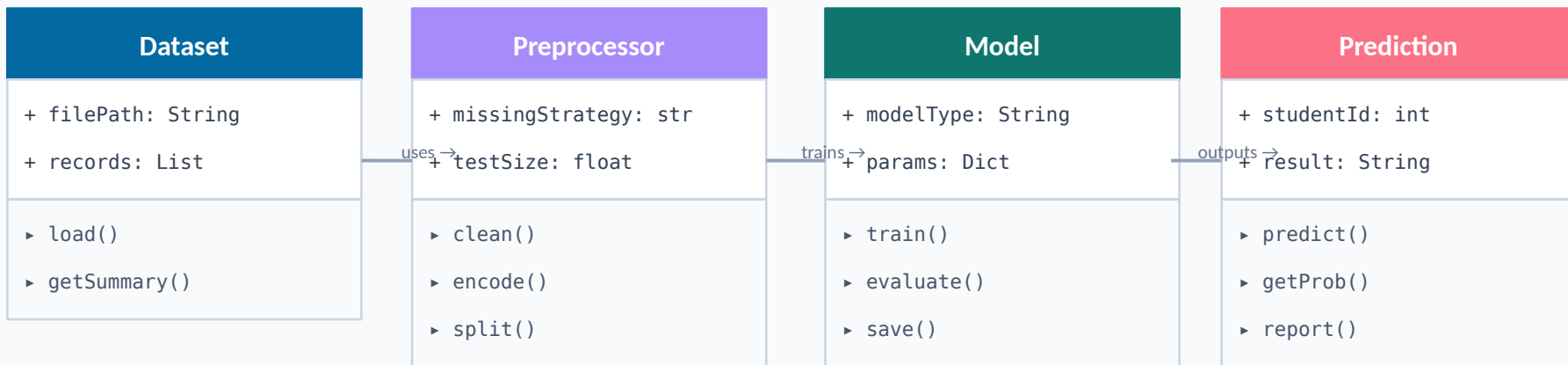
Train-Test Split (80/20)

Randomly split the cleaned dataset: 80% for model training and 20% for unbiased performance evaluation.

Actors & System Interactions



Object-Oriented Structure



Dataset → Preprocessor

Association — Dataset is passed to preprocessor for cleaning

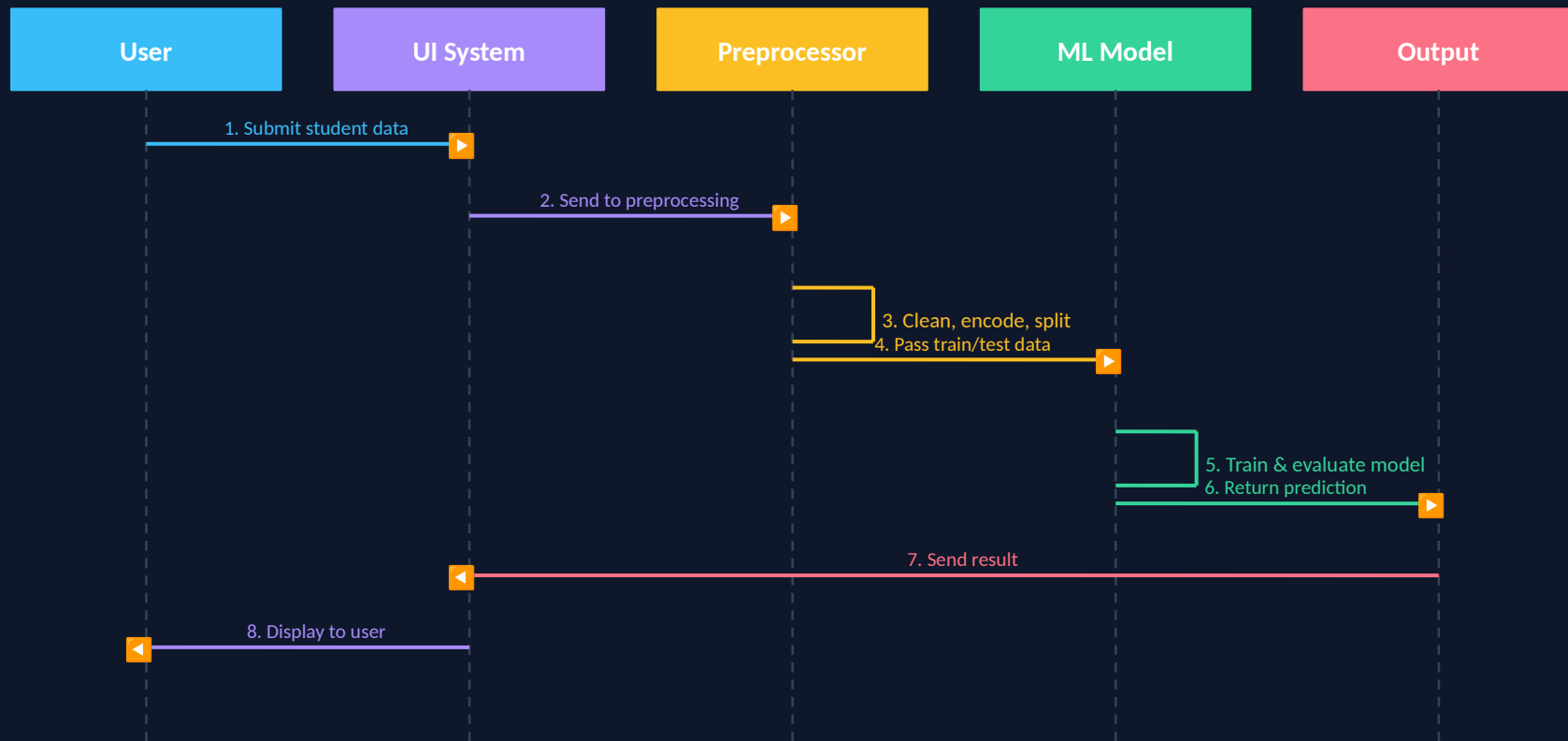
Preprocessor → Model

Dependency — Cleaned data feeds the training pipeline

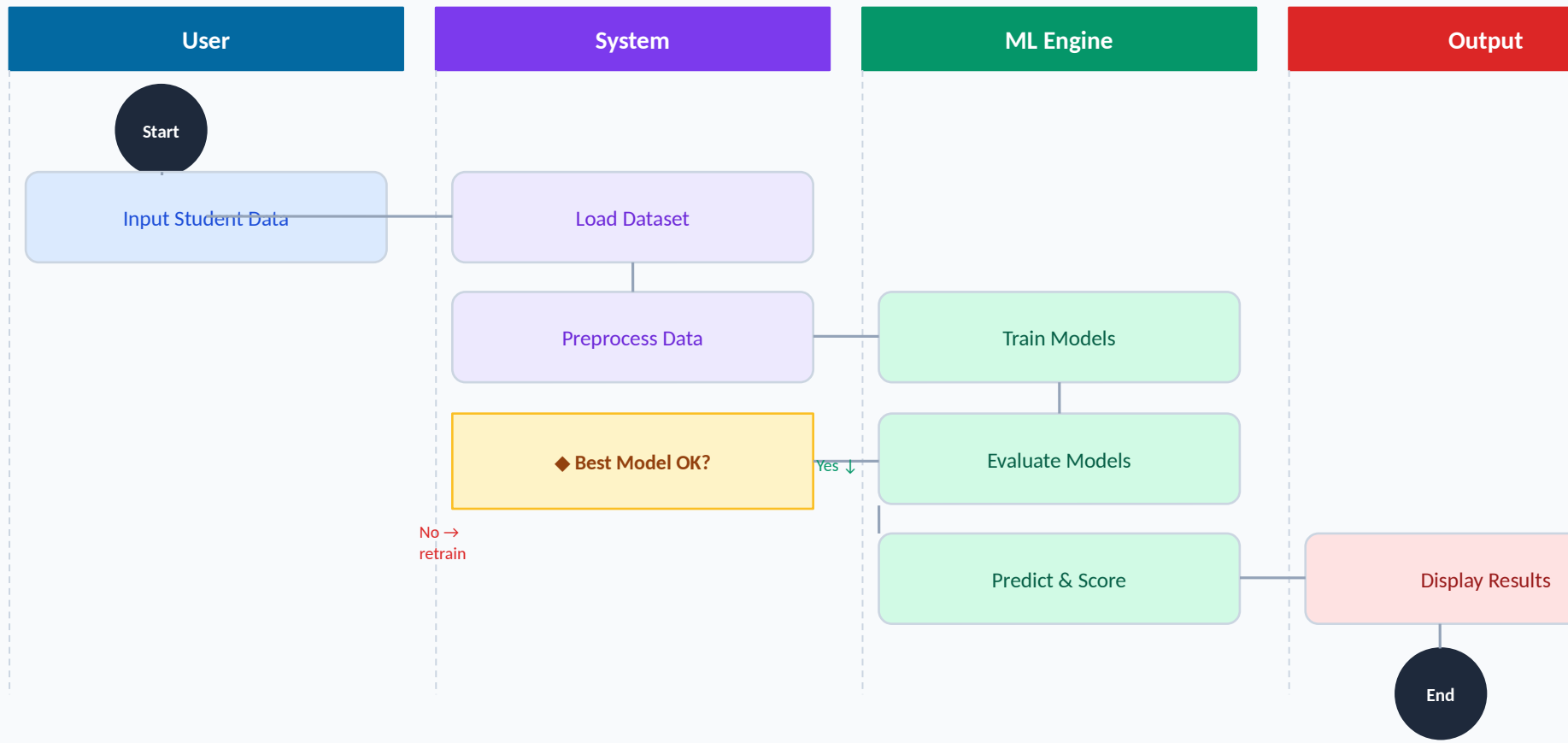
Model → Prediction

Realization — Model outputs a Prediction instance

System Interaction Flow



ML Pipeline Workflow



Model Evaluation Metrics

★ BEST MODEL

Decision Tree	Logistic Regression	Linear Regression
Classification	Classification	Regression
Accuracy	Accuracy	R ² Score
83%	83%	0.98
F1 Score	F1 Score	MSE
0.82	0.83	0.032
Precision	Precision	RMSE
84%	83%	0.179
Recall	Recall	
83%	83%	

★ Linear Regression is the best performing model with R² = 0.98 | Classification models achieve 83% accuracy

Key Takeaways



UML Bridges Gap

UML diagrams successfully bridge the gap between ML engineering and software design, providing a standardized visual language.



Strong ML Performance

Decision Tree & Logistic Regression achieved 83% accuracy; Linear Regression attained $R^2=0.98$ for salary prediction.



Design Clarity

Use-case, class, sequence, and activity diagrams enhanced team communication and system understanding throughout development.



Real-World Impact

The system provides actionable placement probability scores and salary estimates to help students and institutions make data-driven decisions.